THE CLASS OF THE TOTAL SPACE OF A SUBBUNDLE AS THE CAP OF A TOP CHERN CLASS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The bivariant top Chern class attached to a finite locally free module. Same chapter chow.tex, label remark-top-chern-class, invoked at proof line 8861. Fixes what the class c_top of the quotient module is and justifies verifying the formula first for modules of constant rank.
- The geometric case of the top Chern class for a regular section. Same chapter chow.tex, label lemma-top-chern-class, invoked at proof line 8892. Supplies the identification of the class of the vanishing scheme of a section cut out by a regular sequence with the cap of the top Chern class, which is exactly the formula for a single integral class.

Source reference. Original lemma 0FA9: sources/stacks-chow/pinned-original-source.tex, source lines 8750–8750. Navigation pointer for \ref{lemma-top-chern-class} used inside the counted span (line 8892); the argument that consumes it is retained in the span.

Source reference. Original remark 0FA4: sources/stacks-chow/pinned-original-source.tex, source lines 7677–7677. Navigation pointer for \ref{remark-top-chern-class} used inside the counted span (line 8862); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup}

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

used inside the counted span (line 8840); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be a scheme locally of finite type over S. Let*

$$0 \to \mathcal{N}' \to \mathcal{N} \to \mathcal{E} \to 0$$

be a short exact sequence of finite locally free $\mathcal{O}_X$-modules. Consider the closed embedding

$$i : N' = \underline{\mathrm{Spec}}_X(Sym((\mathcal{N}')^\vee)) \longrightarrow N = \underline{\mathrm{Spec}}_X(Sym(\mathcal{N}^\vee))$$

For $\alpha \in \mathrm{CH}_k(X)$ we have

$$i_*(p')^*\alpha = p^*(c_{top}(\mathcal{E}) \cap \alpha)$$

where $p' : N' \to X$ and $p : N \to X$ are the structure morphisms.

Proof. Here $c_{top}(\mathcal{E})$ is the bivariant class defined in Remark 0FA4. By its very definition, in order to verify the formula, we may assume that $\mathcal{E}$ has constant rank. We may similarly assume $\mathcal{N}'$ and $\mathcal{N}$ have constant ranks, say r' and r, so $\mathcal{E}$ has rank $r - r'$ and $c_{top}(\mathcal{E}) = c_{r-r'}(\mathcal{E})$. Observe that $p^*\mathcal{E}$ has a canonical section

$$s \in \Gamma(N, p^*\mathcal{E}) = \Gamma(X, p_*p^*\mathcal{E}) = \Gamma(X, \mathcal{E} \otimes_{\mathcal{O}_X} \mathrm{Sym}(\mathcal{N}^\vee) \supset \Gamma(X, \mathcal{H}om(\mathcal{N}, \mathcal{E}))$$

corresponding to the surjection $\mathcal{N} \to \mathcal{E}$ given in the statement of the lemma. The vanishing scheme of this section is exactly $N' \subset N$. Let $Y \subset X$ be an integral closed subscheme of δ-dimension n. Then we have

(1) $p^*[Y] = [p^{-1}(Y)]$ since $p^{-1}(Y)$ is integral of δ-dimension $n + r$,
(2) $(p')^*[Y] = [(p')^{-1}(Y)]$ since $(p')^{-1}(Y)$ is integral of δ-dimension $n + r'$,
(3) the restriction of s to $p^{-1}Y$ has vanishing scheme $(p')^{-1}Y$ and the closed immersion $(p')^{-1}Y \to p^{-1}Y$ is a regular immersion (locally cut out by a regular sequence).

We conclude that

$$(p')^*[Y] = c_{r-r'}(p^*\mathcal{E}) \cap p^*[Y] \quad \text{in} \quad \mathrm{CH}_*(N)$$

by Lemma 0FA9. This proves the lemma. □